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Evaluating the effects of collaborative governance: Case of a digital education project

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ABSTRACT

The introduction of digital curricula has become topical due to the increasingly pervasive role of ICTs in the society implying the necessity to develop digital skills from a young age. Because of the lack of in-house know-how, governments often need to rely on external actors providing digital education expertise. Organisational structures of such programs thus resemble collaborative governance regimes with distinct roles attributed to specific stakeholders. In this study, we explore cross-sector collaboration structures of the DigEd project aiming to introduce digital education in different cycles of compulsory and post-compulsory schooling on the local level in Switzerland. Findings indicate that the lack of leadership and of clarification of each partner's role during the pilot phase of the project did not allow for the creation of a shared understanding among the involved partners and led to the domination of a limited number of influential actors. This tendency is confirmed through a graphic SNA analysis clearly illustrating the centrality of non-state actors. This contribution shows that it is not only the effects of projects that should be of interest to evaluators, but also their internal collaboration structures.

1. Introduction

Collaborative approaches to governance have become indispensable for developing public policies addressing complex issues requiring sharing data, technology, and other resources (Gasco-Hernandez et al., 2022; Sørensen & Torfing, 2021; Woodland & Hutton, 2012). The multiplication of scientific publications in the last decade speaks to the proliferation of such questions and to the necessity to solve them in innovative ways implicating both state and non-state stakeholders (Bianchi et al., 2021; Gil-Garcia et al., 2019; Hysing, 2022; Yang, 2023). The introduction of digital curricula in public education systems falls under this scope since governments tend to not have at their disposal the necessary know-how for the successful implementation of such programs. Consequently, the reliance on external partners regarding the provision of digital education expertise may be necessary. At the same time, the mastery of digital skills conditions the ability to navigate in the modern society and determines future career prospects of young people (Elphick, 2018; Otterborn et al., 2019).

Although commonplace for other policy fields, such as those related to environmental and health issues (Guerrero et al., 2015; Lu et al., 2022; Robinson et al., 2020; Tremblay et al., 2019), the evaluation of collaborative structures in the framework of digital education projects is a novel topic with the introduction of digital curricula taking place only since recently. At the same time, evaluating collaborations has been deemed necessary due to the high likelihood of them being "challenging and limited" (Greenwald & Zukoski, 2018, p. 323), successful collaboration being difficult to achieve and failure being prevalent (Hughes & Weiss, 2007; Marek et al., 2015). In this study, we analyse the collaborative structures of a digital education (DigEd) project initiated by a Swiss local government with the objective of introducing the teaching of digital skills in different cycles of compulsory and post-compulsory schooling. The project regroups different actors involved in the development of digital education on the local level. Our key ambition is to explore the role of the project's main stakeholders and their impact on the quality of collaboration and achievement of the project's objectives. We accomplish this objective with the use of the social network analysis (SNA) allowing us to explore links between key partners and collaborators of the project.

Basing ourselves on the Bryson, Crosby and Stone's (2015) model of cross-sector governance (Fig. 1), we characterise collaboration as an evolutive process composed of distinct stages and specific determinants interacting either chronologically or non-sequentially. Bryson et al. (2015, p.648) consider "cross-sector collaboration as the linking of information, resources, activities, and capabilities by organisations in two

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or more sectors to achieve jointly an outcome that could not be achieved by organisations in one sector separately". Through their review of theoretical frameworks and empirical works expanding over a decade, Bryson, Crosby and Stone provide a thorough explanation of multi-faceted elements influencing collaborative dynamics. Additionally, we subscribe to an ongoing examination of collaborative dialogue, decision-making and shared purpose constituting cornerstones of a successful collaboration (Woodland & Hutton, 2012).

With this framework in mind, we present the results of a survey dispatched to all collaborators of the DigEd project. The interplay of institutional, organisational and individual aspects of collaboration that we uncover allows us to paint a complete picture of the project's governance. We focus on the exploration of the evaluation of collaborative processes since we cannot yet judge whether the objectives of the ongoing project have been attained. By situating the evaluation of collaborative structures in the specific context of the DigEd project, we contribute to the literature in two ways. Firstly, given that our data reflect the perceptions of frontline workers and managers, we respond to the call for more research studying other than top management levels of collaborative projects' hierarchies (Gasco-Hernandez et al., 2022; Woodland & Hutton, 2012). Secondly, we link the conceptualisation of collaborative governance with continuous project evaluation. In our study, the quality of collaboration impacts directly on the achievement of policy objectives. Given that our evaluation strategy is based on an evaluation plan with precise measures and indicators, we respond also to the previous call for rigorous collaborative governance evaluation

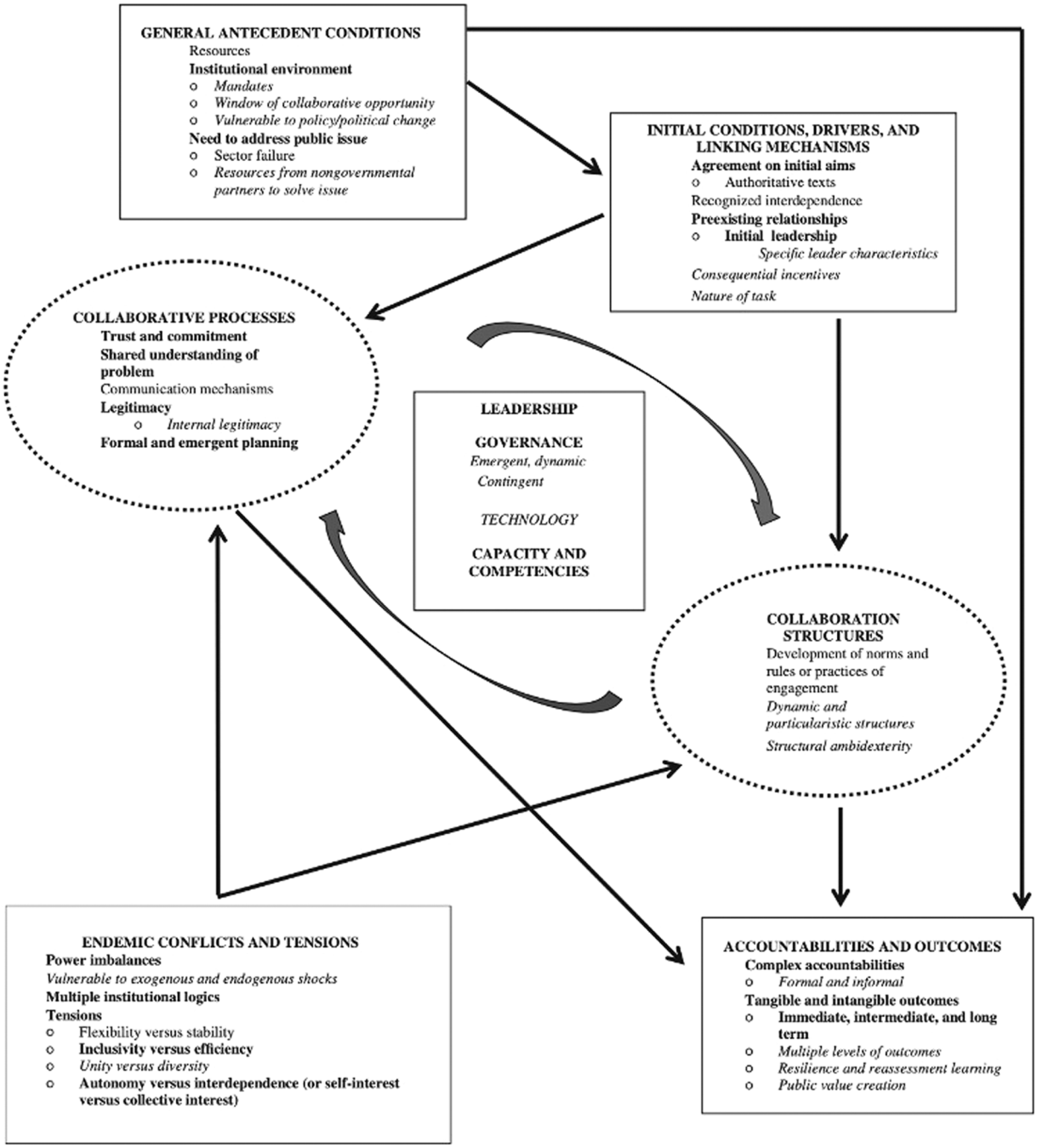

Fig. 1. The Bryson, Crosby and Stone summary model of major theoretical frameworks and empirical studies (2006–2015) on cross-sector governance (Bryson et al., 2015).

(Voets et al., 2021).

This article consists of five sections. In the following section, we conceptualise and typify collaborative governance, as well as present recent findings on the evaluation of collaborative governance and the most important factors having an impact on its conduct. In the third section, we discuss our case study and methodological approach. In the fourth section, we present our findings based on the results of a survey distributed to all collaborators of the DigEd project and an SNA model identifying the most influential stakeholders of the project. In the fifth section, we conclude by pondering the wider importance of our findings for the field of collaborative governance evaluation.

2. Evaluation of collaborative governance

Government-led digitalisation projects often require the involvement of non-state actors for their successful development and implementation (Breaugh et al., 2023). Such collaborative regimes can increase the support of public policies and facilitate their implementation through the involvement of actors with a better understanding of local contexts (Calò et al., 2024; Fliervoet et al., 2016; Hysing, 2022; Lehtonen & Uusikylä, 2021). Moreover, they have been used for addressing complex problems exceeding the capacity of the State and thus allowing to "create public value that could otherwise not be achieved" (Fliervoet et al., 2016; Sørensen & Torfing, 2021; Voets et al., 2021).

The evaluation of collaborations in the framework of public policy development and implementation has intensified in the recent years with the multiplication of collaborative settings and the abundance of encountered challenges (Fliervoet et al., 2016; Lehtonen & Uusikylä, 2021; Robinson et al., 2020). The evaluative processes have been considered crucial due to the possibility to "reveal critical junctures in the collaborative process that enhance or impede project developments" (Breaugh et al., 2023, p. 2). Their objectives have revolved around the investigation of how collaborative processes should be organised to support innovation and public value creation (Lopes & Farias, 2022). The existing research shows that external and internal conditions characterising collaborations are critical in this regard and generate drivers of and barriers to collaborative governance (Ansell & Gash, 2008; Bryson et al., 2015; Emerson et al., 2012).

Collaborative governance regimes can be found in different policy sectors, involve different types of actors and play by different rules. In their 2003 article, Mandell and Steelman discussed "interorganizational innovation" referring to organisational arrangements between public, private and non-profit organisations (Mandell & Steelman, 2003). The authors distinguished between five types of collaboration based on the intensity of interaction, commitment and resource-sharing of involved organisations. A few years later, Agranoff (2007) typified four types of collaborative networks based on their purpose: informational, developmental, outreach and action. In informational networks, involved actors exchange information and decide themselves on taking action. Under developmental arrangements, the roles of actors are limited to the provision of training, research or other services. The purpose of outreach networks consists in elaborating strategies for change, involving an exchange or coordination of resources. However, decision-making and implementation still stay within the competence of actors. Finally, action networks adopt collaborative strategies and develop and deliver services in a coordinated way (Agranoff, 2007; Estermann et al., 2009).

Perceiving the importance of governance structure that had been missing in the previous typologies, Diaz-Kope and Miller-Stevens (2015) distinguished between three types of collaborative governance structures based on the variable of governance: interagency governance, cross-sector governance and grassroots governance. The interagency governance structure is characterised by the dominant role of government and its agencies in decision-making and resource provision with a variable reliance on community trust, social capital and private sector knowledge. The cross-sector governance structure is defined by a reliance on financial, technical and human capital from government entities and by a reliance on community resources and social capital. The decision-making processes are conducted in cooperation with all involved state and non-state actors. The grassroots governance structure is characterised by its minimal reliance on government funding and by a heavy reliance on community resources and social capital. Decision-making is conducted primarily independently of government agencies (Diaz-Kope & Miller-Stevens, 2015).

In parallel to the typology of collaborative governance, its definition has also evolved over the last decades with its tenets being the presence of multiple public agencies and non-state stakeholders engaging in formal and consensus-oriented decision-making processes aiming to make or implement a public policy (Ansell & Gash, 2008; Bingham, 2011; Emerson et al., 2012). The Ansell and Gash's model (2008) constituted one of the first attempts at conceptualising the dynamics of collaborations. The two authors related the starting conditions and collaborative processes with the outcomes of collaborative governance. Several years later, Emerson et al. (2012) added the external context of collaborations to the equation in their collaborative governance framework. Building on the previous research, the summary model of cross-sector governance coined by Bryson et al. (2015) delves further into the relation between different contextual elements and their mutual influence. It thus represents the most comprehensive conceptualisation of collaborative governance.

Through the review of numerous empirical works and theoretical frameworks, Bryson et al. (2015) present a model illustrating the main topics found in cross-sector governance (Fig. 1). Their model indicates which elements have the most importance in the field of collaborative governance research and how they are interconnected and influence each other. Therefore, studies have proven that institutional and environmental conditions, collaborative processes and structures, history between the actors or the assessment of the collaboration's outcomes are the main aspects that define the cross-sector governance context. As such, they may also constitute the key elements that need to be explored by the evaluators of collaborative regimes.

Given the decisive impact of governance structures on the quality of collaboration, joint decision-making processes conducted by both state and non-state actors, the reliance of government agencies on the expertise of non-state actors and the reliance of non-state actors on government's resources, we typify our case as a cross-sector governance structure based on the typology of Diaz-Kope and Miller-Stevens (2015). With this classification in mind, we adopt, in the following sections of the article, the Bryson, Crosby and Stone's (2015) model of cross-sector governance for the study of the DigEd collaborative regime.

Regarding the institutional environment, Bryson et al. (2015) advance that a clear understanding of a collaboration's objectives is necessary to define each partner's role – this often comes in the form of a written convention. The clarity of roles and responsibilities of involved parties, as well as inter-institutional information sharing have been deemed particularly important for building trust among partners and increasing their willingness to participate (Gil-Garcia et al., 2019; Margerum, 2001). On the individual level, the clarity of roles contributes to increased job satisfaction and commitment of organisational members. At the same time, it reduces tension and job-related stress (Gil-Garcia et al., 2019). "Having a clear sense of what should be done to achieve common goals in a collaborative effort gives participants confidence about what they need to do and what they can expect from other participants" (Gil-Garcia et al., 2019, p. 2). The understanding of roles and responsibilities is facilitated using boundary objects; collaboration, coordination and communication skills; and the diversity of participating organisations (Sayogo et al., 2016). Boundary objects can take the form of regular meetings and implementation plans related to collaboration structures. Moreover, written documentation formalising the roles and responsibilities of participating actors can validate the attributed roles and responsibilities and increase their acceptance. Considering diverse interests and constraints of involved parties further contributes to the collaboration's success (Gil-Garcia et al., 2019).

On the organisational level, Bryson et al. (2015) highlight the impact of leadership on the collaborative environment, especially its structures and processes. Having clearly designated leaders is crucial as their goal is to unify the different parties while underlining each partner's specific contributions and clearing any tensions that could arise between them. The key role of sponsors and champions of collaborative projects for their success is undisputed in the literature (Agranoff, 2012; Calò et al., 2024; Crosby & Bryson, 2010). More recently, Lopes and Farias (2022) conducted an extensive literature review of organisational factors facilitating and hindering collaborative innovation processes in the public sector. They found that internal characteristics of organisations, organisational relationships, tools and strategies for citizen participation and managerial roles and leadership practices all played a role in their success. On this topic, Breaugh et al. (2023) have found that the establishment of rules, project structures and standard operating procedures had been important especially in the starting phase of collaborative projects. In later phases, leadership skills became decisive for handling problems and moving toward implementation (Breaugh et al., 2023). Studying collaborative governance in a homelessness policy in Canada, Doberstein (2016) found that "strategic use of rules and steering by bureaucrats keeps the ship on course while harnessing the dynamism of collaborative governance" (p. 838).

Van Dijck and Steen (2022) focused on the investigation of the role of political support in the framework of collaborative projects and found that whereas the role of leadership has been evaluated overwhelmingly positively in previous studies, the impact of political support and of organisational culture could be positive or negative depending on context. Political support can act as a double-edged sword in collaborative governance processes. On the one hand, politicians can provide funding to a project, prioritise it above other policy initiatives, connect stakeholders, resolve conflicts and increase collaboration's legitimacy. On the other hand, however, political pressure can also lead to the confusion of policy objectives, loss of politically neutral status and obstruction of projects (Van Dijck & Steen, 2022). Furthermore, political narratives often change and require public officials to repeatedly translate them into policy actions (Nicklin, 2019). At the same time, the influence of politicians on collaborative projects is often downplayed with its importance being emphasised especially during the initiation phase (Van Dijck & Steen, 2022).

On the individual level, we observe that despite the undisputed importance of effective leadership for the success of collaborative governance structures, the role of lower-level managers and employees stays understudied (Gasco-Hernandez et al., 2022). The rare existing research shows that qualities such as the ability to work across boundaries, involve and engage stakeholders, participate in teamwork (Hartley et al., 2013), and take risks (Mergel, 2018) stimulate collaborative environments. In addition, Social Network Analysis is a useful method to map out interactions between actors, as it can illustrate how a network can have an impact on policy outcomes (Christopoulos, 2008; Fliervoet et al., 2016; Woodland & Hutton, 2012). Incentives to participate are important aspects to take into consideration. Partners might have different expectations and goals in mind, which can result in differing motivations and values. Overall, participation in collaborative projects should have a clear advantage to gain for all involved partners (Bryson et al., 2015; Lopes & Farias, 2022). It is acknowledged that bringing different actors together can often cause tensions due to differences in interests and beliefs (Bodin et al., 2020). Witting et al. (2020: 197) talk about network governance as "a situation in which state actors and non-state actors are equal partners and their action is based on reciprocity and trust". Henry (2011: 316) uses the term of policy networks which "describe patterns of interaction among actors working a particular policy system […] in reference to a particular type of relationship such as information exchange or political coordination". Additionally, policy networks are seen as being key parts of the success or failure of governance systems (Dietz & Henry, 2008). The analysis of network structures can uncover multiple things such as entrenched political conflict and non-cooperation (Henry, 2011) or better capacities for solving cooperation or coordination problems (Bodin et al., 2020).

3. Research design

3.1. Field of study

The digital education project studied here represents an ambitious effort of a Swiss local government to introduce the teaching of digital skills in its compulsory – composed of three distinct school cycles – and post-compulsory school system. The project's objective was initially defined in general terms as "the development of digital and transversal skills of all students with the objective of increasing their emancipation […] to evolve as active and enlightened citizens in the informational society of today and tomorrow" (Caneva, 2019: 12). It began in the school year 2018/2019 in the first schooling cycle and its implementation will progressively continue until 2028 in the subsequent cycles. Pilot phases lasting two to three years for each school cycle precede the generalisation of the project in every school. The pilot phases allow for the testing of different approaches and methods and, if needed, the readjustment of different aspects before the full-scale deployment.

Over the years, the project has gained attention and inspired neighbouring local governments looking to implement digital education in their school systems. Therefore, it has been crucial for the local government to collaborate with people who are sufficiently driven and skilled to deliver a project that could be showcased. Given its important political dimension, the project's conceptualisation has since the beginning involved several stakeholders. On the one hand, we find state authorities responsible for its conception, its organisational aspects and for the delivery of necessary technological equipment to participating schools. On the other hand, we find three academic partners (non-state) specialised in various fields involving pedagogy, information technology and social sciences. Their main responsibilities include the conceptualisation and instruction of digital education training for participating teachers. A public administration institute attached to the local university is responsible for the project's evaluation. Both authors of this article are evaluators of the project in question. To preserve their anonymity, the involved state and non-state actors will not be named. Thus, non-state partner institutions have been assigned labels "Partner A", "Partner B", "Partner C" and "Partner D". Where relevant, the State authority's sub-departments have been assigned labels "State A", "State B" and "State C".

The "raison d'être" of the collaborative governance arrangement in the framework of the DigEd project is related to the digital education being considered as a "wicked problem" exceeding the capacity of the State and necessitating the involvement of non-state actors. "Wicked problems cannot be clustered within single organisational boundaries because they possess dynamic and complex characteristics, involving multi-level, multi-actor, and multi-sectoral challenges'' (Bianchi et al., 2021, p. 1581). In the present case, the local department of education called on external actors to make up for the lack of know-how in the development of teachers' digital competencies for the teaching of digital skills. According to Hysing (2022), such collaborative arrangements fall under the "policy support" schemes. Hysing (2022) divided the strategic purpose of collaborative governance between policy support and voluntary action. Under policy support, he understands collaborations that aim to obtain input on policies from non-state actors with the objective of improving the policy in question or facilitating its implementation. Collaborative governance regimes seeking voluntary action encourage participation of non-state actors to support public policies and increase the effectiveness and efficiency of processes.

The realistic and continuous evaluation of pilot phases of the project has been crucially important for its optimisation and long-term sustainability. At the time of writing, the DigEd project has moved from pilot phase to general deployment in the first and second school cycles and is still in pilot phase in the third cycle and on post-compulsory

levels. The results of the first three evaluation reports published in the summers of 2019, 2020 and 2021 showed that shortcomings related to the project's organisational aspects had impacted greatly on teachers' satisfaction and project's acceptance. The quality of collaborative processes between the project's partners thus became the key factor influencing the achievement of its objectives.

3.2. Methods

We use data collected through a standardised online questionnaire in 2022 in the context of the DigEd project's evaluation. The inclusion of several groups from the target population (e.g., project managers, researchers) in the questionnaire's construction was essential to gain valuable information on the collaborative context (Nichols, 2002). The survey focused on different elements touching on the structure, processes and dynamics in place within each project partner as well as on the general collaborative context. Most survey items had a 6-point "agree-disagree" response scale and were coupled with open-ended questions. The survey was sent to all people with tasks linked to the project on the institutional level. The response rate was 83 % (n = 93) with collected responses representing all project partners. The sample composition depicts a wide demographic variety of the project's population in terms of occupied position, seniority, work rate and partner affiliation. Through the collaborators' personal assessment of the project's different elements, we expect our results to allow us to draw an accurate picture of the collaborative environment and its evolution. In addition, open-ended answers enrich the collected information and help us understand the collaboration's dynamics.

Because we use secondary data for this study, not all aspects from the Bryson (2015) model could be investigated. However, its elements and the related studies discussed in the theoretical section are essential to grasp the dynamics at play in the evaluation of the DigEd project and provide a key to understanding it, especially on the individual level of analysis. Indeed, while SNA provides an insight on the dynamics at play between the partners, its interpretation relies mainly on the theoretical framework.

The data presented in the first part of the findings section evaluates internal communication and organisational structures, as well as intra-organisational culture using scales developed by Bryde (2003) and by Westerveld (2003). Controlling for any major differences between institutions on these elements was important for deciding whether internal aspects influenced the collaborative environment. For the Social Network Analysis part, we considered both sociocentric and egocentric approaches. While the former focuses on ties inside a given social network (Marsden, 2002) – meaning here the relationship between partners – the latter focuses on individuals and their direct contacts inside a network (Witting et al., 2020). Firstly, in the scope of the DigEd project, each respondent was asked to name the three institutions they collaborated with the most outside of their home institution. Secondly, applying the same instruction, the respondents were asked to name three people they had the most connections with. This way, we were able to define the nodes (individuals and partners) and edges (relationship) composing our network. With this mix of common sociocentric and egocentric network data collection methods (Marsden, 2002), we were able to identify high-leverage groups and individuals carrying out the tasks and activities central to the collaboration (Woodland & Hutton, 2012).

For our analysis, we consider the following measurements. Closeness centrality: high values indicate a shorter time for a node to communicate with all other nodes (Li & Huang, 2023). Betweenness centrality: indicates how many times a node is on shortest paths between each pair of nodes in a network (Grandjean, 2021). Density: represents the number of observed edges divided by the number of possible edges (Witting et al., 2020). Average path length: indicates the average number of edges between two nodes on a given distance for all the network's paths. It comes as a good complement to density which can be highly affected by remote nodes (Grandjean, 2021). Clustering coefficient: represents the tendency of nodes clustering together and the "presence of a more localised density" (Grandjean, 2021: 16). The accumulation of high degrees, high betweenness and/or closeness centralities allows us to identify a network's core nodes (Li & Huang, 2023). Average path length and average clustering coefficient define a network's cohesiveness (Li & Huang, 2023). Thus, higher path lengths and lower clustering coefficients make for a less cohesive network.

The network graphs were computed with the open-source software *Gephi* (v 0.10) using the *ForceAtlas* layout "made to spatialise small-world networks" (Gephi, 2011: 11) allowing for better interpretation and good readability in the present case. As Henry (2011: 363) argues, "it is the job of the analyst to attribute theoretical significant meanings to the [edges] and the nodes [due to their abstract mathematical nature]". Therefore, given its complexity and evolution over time, understanding the context of our network is central to explain and give a meaning to the SNA graphs. The results presented in the following section first aim to bring out the main elements affecting the collaborative environment by exploring internal project management aspects. These results will then bring interpretative elements to the second section showing the SNA graphs generated for both sociocentric and egocentric approaches. By looking into the relationships inside the project's network and using our personal field knowledge, we aim to understand how issues have risen and persisted over time.

4. Findings

4.1. Internal project management aspects of the DigEd project

Overall results show that respondents evaluate positively the internal project management aspects presented in the questionnaire. Most collaborators self-assess the internal organisation of their partner institution of affiliation as satisfactory, efficient, coordinated and open to dialogue.

However, we observe elements assessed with less positivity and showing differences between institutions. We find a fairly low degree of agreement[1] on the sufficiency of project-related resources (financial and equipment) among Partner C (53.9 %, n = 13) and State respondents (52 %, n = 25) compared to the two other partners (Partner B 90 %, n = 30; Partner A 87.5 %, n = 24). These results reflect an ongoing problem concerning the implication of Partner C in the project which was informally defined for a long time. Most resources have thus been allocated to the other two non-state partners. The State partner being divided into multiple sub-departments; these are often composed of only a few people simultaneously in charge of multiple projects. This results in resources running short when tasks must be done in haste.

A similar breakdown between institutions is observed regarding the clarity of roles and the definition of the collaborators' responsibilities in the project. While these aspects are overall evaluated as sufficient, this tendency is again less marked with the State and Partner C. Partner B (80 %, n = 30) and Partner A respondents (79 %, n = 24) largely agree on the clarity of roles and internal definition of responsibilities. This is less the case for Partner C (61.5 %, n = 13) and even less for the State (48 %, n = 25). Assessing the regularity of information communication about the project, we observe a high level of agreement among Partner B (80 %, n = 30). This rate is lower among Partner A (62.5 %, n = 24) and Partner C (52.9 %, n = 13) while is it at the lowest among State respondents (40 %, n = 25).

Important issues regarding procedures in place as well as task-related deadlines are also observed. Overall, less than half of respondents

[1] By degree of agreement, we aggregate the three "agree" answer options ("Totally agree", "Agree", "Somewhat agree"). By degree of disagreement, we aggregate the three "disagree" answer options ("Totally disagree", "Disagree", "Somewhat disagree").

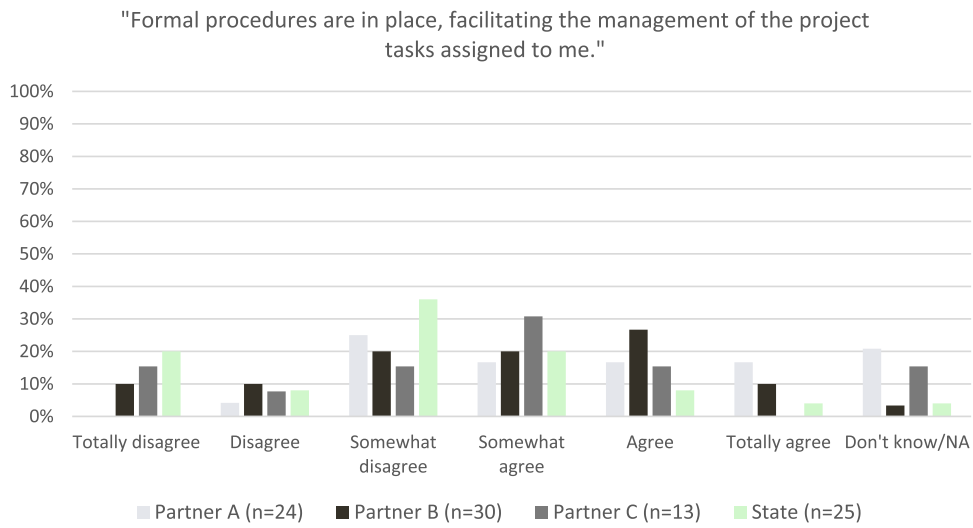

Fig. 2. Proportion of answers by institution.

(47.3 %, n = 93) feel that facilitating processes for the management of project tasks are lacking (Fig. 2). We see differences in opinion on this aspect between the academic partners - Partner B (56.7 %, n = 30), Partner A (50 %, n = 24) and Partner C (46.1 %, n = 13) - and the State (32 %, n = 25). On the assessment of imposed deadlines (Fig. 3), the split is even clearer between the academic partners. While the level of agreement regarding ease to respect deadlines for Partner B is the highest (56.7 %, n = 30), it is lower for the other non-state partners (Partner A 45.8 %, n = 24; Partner C 30.8 %, n = 13) and extremely low for State respondents (8 %, n = 25).

The analysis of open question answers allows us to advance that the different results between partners and negative assessments can be attributed to the lack of a clear collaborative convention. This absence of a written convention and power asymmetries between the different partners are often quoted by respondents as the main causes of their negative assessments. Coordination and efficiency are consequently affected, making it hard for some teams to put in place finer project management procedures. With an unclear vision of the objectives, roles and responsibilities of partners and collaborators, the latter are concerned about its impact on collaborative processes. The scarcity of project-related information affects internal communication about the project, consequently leaving some collaborators with a low level of anticipation for conducting certain tasks. Some people therefore explain going through stress-fuelled situations due to last minute deadlines or conflicts. Unsurprisingly, this situation creates a bellicose climate on the collaborative level, between partners and individuals. The project's recent change in governance structure (i.e., project managers group) is a direct attempt to address this.

Focusing on this new structure, 43.5 % (n = 92) of respondents feel it has improved their work situation and the collaborative environment. More than half of the surveyed collaborators (54.3 %, n = 92) declared being satisfied with these recent changes while only a small minority declared being unsatisfied (7.6 %, n = 92). Thus, while it appears that changes in leadership are relatively welcomed by respondents, its recency at the time of the survey resulted in more than a third of people unable to express an opinion on the matter. Therefore, a clear assessment of the new governance's impact on the collaborative context is difficult.

Stretching across multiple school cycles, the project follows an asymmetric evolution and has encountered multiples issues over the years. However, efforts to rectify things in the third school cycle have been made. A fair amount of survey respondents praises the organisational process around this cycle, allegedly more inclusive and collaborative with a dynamic dialogue that considers every partner's point of view. Most respondents insist on separating negative governance elements from the organisation around the third cycle where the actors feel like their competencies are really integrated and the result reflects their combined efforts.

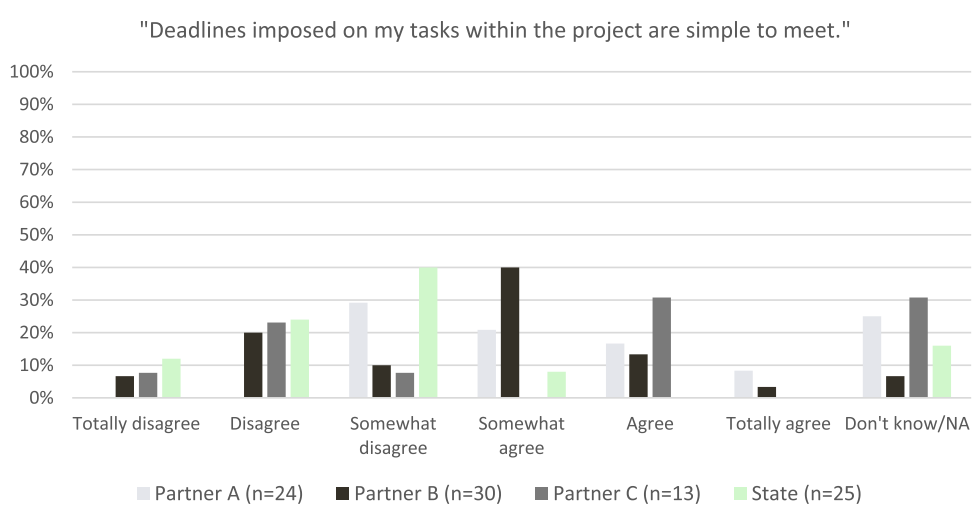

Fig. 3. Proportion of answers by institution.

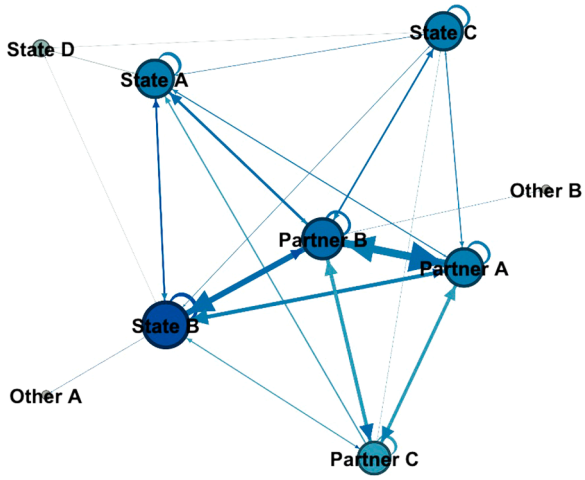

Fig. 4. SNA (sociocentric) representation of the connections between project partners.

4.2. Social network analysis of the DigEd project

The following graph (Fig. 4) depicts the interactions between the project partners. These are based on individual interactions on project-related topics (e.g., research, administration, strategic discussions) declared by respondents (n = 91). The bigger an edge or a node is, the more connections it has. We identify the following core nodes based on degree, betweenness and closeness centralities values: State B (15, 13.67, 0.89), Partner B (13, 6.67, 0.8) and Partner A (12, 0.67, 0.73). This is representative of their size in the project (i.e., number of collaborators) and of their level of interaction in the context of teacher training and educational content. Partner B's out-going edges towards the state partners appear heavier than in-going edges meaning visibly more frequent interactions in comparison to the other non-state institutes such as Partner C. With an average path length of 1.41, graph density of 0.65 and clustering coefficient of 0.63, the network has a high cohesion. State A and State C show a relatively high betweenness centrality value (1) in this network, proving their importance as state actors. The centrality of Partner B is quite evident in the network. In comparison, Partner C scores low on most values measured here, depicting its lack of influence/implication in the network despite having strong ties with the other partners.

Fig. 5 depicts the egocentric network of the project mapping out the individual connections between collaborators outside of their own institutions. Each respondent indicated the three people they had the most project-related interactions with (1 person, n = 72; 2 people, n = 69; 3 people, n = 56). Nodes represent collaborators and are coloured according to their affiliation (names are omitted here for privacy reasons). Due to the low number of personnel in certain sub-departments, all State collaborators have been regrouped under the same label. The thicker edges are a visual representation of mutually confirmed links. The network presented in Fig. 5 has an average path length of 4.02, a graph density of 0.04 and a clustering coefficient of 0.25. Compared to the sociocentric graph (Fig. 5), we see that these values are much lower on the egocentric level. A lower density means fewer intensive interactions between nodes and a higher dependency of a node on another node to relay information (Witting et al., 2020; Li & Huang, 2023), while a lower clustering coefficient and higher path lengths signify less cohesion in the network (Li & Huang, 2023). This is apparent when looking at the number of nodes with only one edge (dyads) or that are the starting point (hubs)/the ending point (authorities) of many edges. Presumably, these hubs are the only ones who can relay information to the other nodes they connect with.

The core nodes (based on degree and betweenness[2]) in this graph are the following: 1. Partner B (15, 1182); 2. State (14, 966); 3. State (15, 502); 4. Partner B (11, 671). The cluster around nodes 1 and 3 represents the higher hierarchy of the project. Node 2 represents the project manager cluster. Additionally, nodes 1 and 2 personify the existing clash between "the old and the new" governance models. While node 1 is the most important in the network, node 2 has shown rapid growth by positioning itself quickly amidst the centre of the network. Node 4 represents the main scientific cluster in charge of developing digital education content.

The different clusters illustrate the way the project functions. Indeed, most collaborators work in groups with specific tasks such as creating content for the different school cycles or coordinating the project. Individuals present from the start of the project show higher betweenness centralities which is expected due to their implication and the ties they have created over time. However, we also see that the project manager group (i.e. the nodes sharing thicker edges with node 2) have many connections with the different parts of the graph. Given the State project managers' recent inclusion in the project, they managed to gain a considerable centrality in the project over a short time-period and in an environment marked by a way of functioning that has been in place for several years.

Many nodes related to Partner B have strong ties with State nodes and compose most of the hubs/authorities in the graph. This illustrates their influence on the project and their leading position, making them the main vector of information between the network's clusters. While this is expected in the context of collaborative governance – non-state collaborators play an important role due to their expertise – and can demonstrate a good level of cooperation, it is problematic when considering the tension and personal conflicts taking place in the DigEd project. In a cooperative context, these central positions can facilitate the flow of information across the whole network. The good functioning of collaborations strongly relies on these central individuals because there are often no alternative routes. In a conflictual context, central positions are powerful and can result in gatekeeping, thus easily cutting off the information flow and favouring non-cooperation (Henry, 2011). When hubs/authorities do not transmit information, a whole cluster of the network is isolated. The results related to the internal management aspects confirm this by showing that conflict between institutions affects communication around objectives, procedures and the time management of tasks.

Betweenness score illustrates well the influence a single node can hold over an entire network. It is particularly useful to highlight brokers or gatekeepers who "[tend] to have opportunities to access information and control [the network's] flowing" (Li & Huang, 2023: 40). Betweenness, in this context, can be interpreted as centralisation (Witting et al., 2020). Thus, the higher the score, the more central an individual is. Node 1, affiliated to Partner B, scores by far the highest value (1182), making it the main bridge between each cluster on the graph and giving it a better position to engage and negotiate in governance processes (Li & Huang, 2023). The second highest score (966) belongs to a State project manager (node 2), demonstrating their reach across the network. The third highest score (857) represents Partner A (node 5), showing its implication in the project through the conception of teacher training content. We note that even though the State is the initiator of the project, it does not occupy the most central positions in the network here.

To conclude the SNA analysis, we have seen that the governance issues of the DigEd project impact on a great number of collaboration aspects The lack of a written convention between partners has generated

[2] Closeness being irrelevant in an undirected graph, betweenness is used instead (Marsden, 2002).

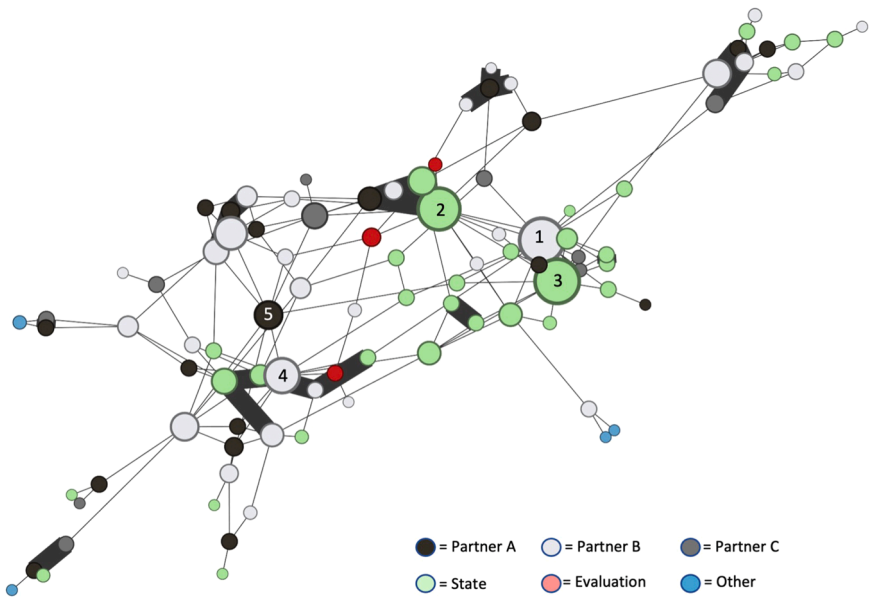

Fig. 5. SNA (egocentric) representation of the connections between project collaborators.

numerous uncertainties concerning collaborative structures and procedures. The lack of a clear leadership makes regulating such aspects a difficult task to conduct. In turn, conflicts and tensions between partners and individuals affect the day-to-day work. Additionally, the implementation of a new governance model (i.e., project managers' group) and a new way of functioning around the third school cycle exacerbates the conflictual situation between some actors since it bypasses their central position in the network as illustrated in Fig. 5. However, we see that the change in governance is seen positively by project's collaborators. The new project managers' position in the network is already very central despite key positions still being held by other actors (e.g., node 1 in Fig. 5). This illustrates the importance of inclusive leadership in such a collaborative environment and its impact on all aspects of the collaboration.

5. Conclusions

Governments around the world find themselves in situations where they cannot develop and implement public policies without the help of non-state actors. The blame is to be put on the increasing complexity of issues necessitating the involvement of external stakeholders. While inevitable, these collaborations need to be institutionalised and follow certain rules and a common purpose. They also need to be continuously analysed and optimised because, as the literature shows, their governance is by no means straightforward and encounters numerous pitfalls. The success of a collaboration depends on both formal (rules, procedures) and informal (facilitative leadership, trust, commitment, shared understanding) factors (Bianchi et al., 2021; Bryson et al., 2015; Emerson & Nabatchi, 2015). Furthermore, the production of positive effects of collaborations may take time and effort as collaborative governance regimes are often also learning processes (Bianchi et al., 2021). In this study, we explored the evaluation of collaborative governance in the framework of the DigEd project, an ambitious initiative aiming to introduce the teaching of digital skills in a local education system in Switzerland. Our data paints a picture of DigEd's inter- and intra-institutional collaborative structures from the point of view of the project's collaborators following the model of theoretical and empirical frameworks of collaborative governance coined by Bryson et al. (2015).

On the organisational level, the DigEd's leadership has been officially entrusted to the State since the beginning. However, for a long time, it was plagued by a reluctance to make and implement any decisions. It thus failed to fulfil its purpose: effectively manage the partnership, build trust among stakeholders (Wang & Ran, 2021), create collaborative spirit and give voice to participants (Ansell & Gash, 2008). The role of leadership for the success of collaborative arrangements has been undisputed in the literature. With effective collaboration being "(…) seriously constrained by a lack of leadership" (Ansell & Gash, 2008, p. 555), it may even constitute its key aspect. Our findings show that the ineffective leadership in the DigEd project was principally found problematic due to the unclear responsibilities of each partner. The lack of a written convention defining the tasks of each partner is linked to several other shortcomings reported by the respondents. Namely, it has had a negative impact on inter- and intra-institutional communication, task-related deadlines and accountability aspects. This finding is aligned with the study of Calò et al. (2024); the authors argue that the establishment of formal agreements framing collaborative relationships leads to reduced transaction costs and make the relationships more stable and predictable. Partners A and B, responsible for the organisation of digital education teacher training, are less dissatisfied with this aspect. We suppose that this result can be explained because both actors had a substantial autonomy in the accomplishment of their tasks and therefore did not perceive the need to clarify their role. In addition, self-assessment from Partner B team was sometimes more positive, possibly because of a stronger political trust and support it received, creating a seemingly more stable work environment. In turn, this asymmetry reflects a power imbalance between partners with more favourable conditions for some than for others. In this connection, the suitability of contextual conditions can also depend on the orientation of the partner in question – some external partners are more concerned with rigor and research and others are more inclined to solve societal problems to stay relevant (Yang, 2023).

On the individual level, the project's ambiguous leadership and objectives and the lack of clear responsibilities of each partner did not allow for the creation of a "we" narrative that would orient the collaboration of participating stakeholders based on shared actions and purpose (Nicklin, 2019). Instead, each of them acted according to their own interests and values. As the results of SNA show, the collaborative structures in the framework of the DigEd project were organised around a limited number of nodes representing the most influential actors deciding about the evolution of the project. Regular and balanced communication between all stakeholders was therefore difficult or

outright impossible. Thus, we assert that the leaders of collaborative governance regimes need to be able to take risks, face adversity and be present on all governance levels. Moreover, to orient their actions, they need to have precise objectives and purpose in mind (Hysing, 2022). Recently, the DigEd collaborative structures and leadership have been strengthened and made more participative by introducing a project managers' group acting as the point of contact between collaborating partners and the project's higher hierarchy. Our survey results show that even though not all respondents have noticed a concrete impact of the new structure on their activities, they perceive its existence positively. The SNA egocentric approach (Fig. 5) illustrates the centrality of the position these project managers have managed to acquire over a short period.

Our findings show the complementarity of survey data measuring the organisational and institutional aspects of collaborative governance based on collaborators' perceptions and SNA data related to the individual level of analysis. In the case of the DigEd project, we see that institutional, organisational and political missteps led to a limited number of dominant actors influencing and determining the collaboration's character. The literature shows us that power and knowledge asymmetries can hinder the commitment of stakeholders when "(…) powerful stakeholders manipulate the process; public agencies lack real commitment to collaboration; and distrust becomes a barrier to good faith negotiation" (Ansell & Gash, 2008: 561). In the DigEd project's framework, a preferential treatment of one non-state actor over the rest of the partners and a one-sided political support toward a specific partner have led to increased tension between stakeholders. In this connection, several authors have found that the diversity of stakeholders was also important for the success of collaborative arrangements (Bianchi et al., 2021; Sayogo et al., 2016). Given that in the case studied here, the three external partners were tasked with providing the State with digital education expertise, we can observe that the diversity of input is not sufficient with the three actors occasionally stepping on each other's toes.

Basing ourselves on the Bryson (2015) framework of cross-sector governance, we studied here the main aspects of a collaborative governance regime in the framework of a local digital education project. Emerging from our findings showing the strengths and weaknesses of collaborative processes on the institutional, organisational and individual levels of analysis, we argue that the evaluation of collaborative structures is crucial and should constitute an integral part of such regimes as their quality can have an important impact on the achievement of policies' objectives. Moreover, such evaluations should be continuous as collaborative structures evolve over time. Despite their inclusion in the existing frameworks, the internal functioning of collaborative partnerships is still largely a "black box" as existing studies are often limited to the analysis of outcomes and outputs of collaborative governance (Calò et al., 2024). The importance of organisational aspects has been accentuated principally in the beginning of collaborations for the purposes of generating a common vision (Lehtonen & Uusikylä, 2021). In this study, we dissected the collaborative structures of the DigEd project and found them to be at the core of multiple issues encountered during its implementation. In particular, we show that power can be concentrated in the hands of a limited number of individuals who thus weaken the collaborative dynamics by becoming the main gatekeepers, decision-makers and information relaters. Gathering data for a SNA in the future would help analyse the evolution of the DigEd collaboration network and observe the effects of the new governance over time. Continuous evaluations allow for the identification of collaboration drivers and barriers. Therefore, they might contribute to increased commitment of stakeholders and lead to the establishment of more realistic and effective collaboration strategies (Margerum, 2001).

This study contributes to the development of knowledge in the field of collaborative governance evaluation that stays under-researched despite the growing interest in the topic in since the beginning of the century. It exposes the main challenges typifying the collaboration of state and non-state actors and emphasises the importance of institutional, organisational and individual aspects of such collaborations for the achievement of their objectives. It thus contributes to the theory and practice of collaborative governance evaluation that has so far insufficiently explored the role of individuals and of their interactions (Kapucu et al., 2017). By doing so, it shows that a continuous evaluation of internal collaborative structures can become an instrument for the further development and improvement of projects. The evaluation of collaborative processes in the field of digital education constitutes a new study area with only a handful of articles dedicated to the topic (Valencia-Tello, 2016). Therefore, our study constitutes one the building stones of such research.

Our study meets a few limitations. For instance, when evaluation results and recommendations are not diffused among partners, their effects are greatly limited. Even when they are known by the actors, there is no guarantee if or when they are taken into consideration and applied. In addition, regarding our case study, using secondary data induced difficulties to address each aspect of the presented models. While the variables selected for the case study covered all dimensions in the Bryson, Crosby and Stone's model, some elements could not be measured. The data collection context was also marked by a certain level of political uncertainty resulting in doubts about the project's future. This might have impacted some respondents' answers, especially since they mostly consisted of self-appraisals. However, textual open-ended answers brought much needed nuance to the data and allowed us to draw a clearer picture of the collaborative dynamics at play.

Furthermore, despite the relevance of its results for the DigEd evaluation, we can note some limitations to the egocentric network representation. Indeed, edges represent solely communication with actors outside of one's own institution and only up to three people could be named. Hence, ties within the same institution are not depicted in the egocentric graph which only shows a portion of interactions for some of the most connected individuals in the network. Likewise, since contact frequency was not measured, it is not possible to define how often contacts happen and if they are recurrent over time. Finally, we lack information about the nature of these interactions in the way that we are unaware if they are on the cooperative or conflictual side. Conflicts can either be at the source of cooperative ties to create different dynamics or create cooperation between conflicted parties to avoid negative impacts for them both (Bodin et al., 2020). In the present context, the project managers (Fig. 5, surrounding node 2) representing the project's new governance are connected to the higher hierarchy (Fig. 5, surrounding node 1) representing the "old" governance. Even though there are disagreements between these two groups, the size of the network makes it difficult for them to not connect with each other. Nonetheless, SNA graphs provide a visual representation of the dynamics in place inside of the DigEd project by mapping out the main interactions between partners and representing the core nodes. Actors who are best placed to facilitate or hinder the information flow between individuals or have the broadest reach across the entire network appear clearly.

Lessons learned

Our research and findings allow us to share the following experiences:

1. The quality of collaborative governance structures has a decisive impact on the commitment of collaborators and achievement of project objectives.
2. The evaluation of organisational and individual aspects should constitute an integral part of evaluation plans and should be conducted continuously during the lifetime of every project based on collaborative governance. Collaborations should be approached as learning processes.
3. The responsibilities of each stakeholder and the leadership of the project should be clearly defined at the beginning of every collaboration. Decisions on collaborative aspects should be taken jointly by

all concerned actors. They should be documented and circulated regularly and transparently among collaborators.

4. The mismanagement of collaborations can lead to delays in a project's deployment and may necessitate the deployment of additional resources for matters such as conflict resolution, accountability attribution and creation of a shared narrative among stakeholders.
5. Collaborative governance structures should not be dominated by a limited circle of actors to prevent information asymmetries undermining the commitment of collaborators. Social Network Analysis constitutes a valuable tool for mapping interactions in a collaborative governance context.

Author statement

Tereza Cahlikova: Conceptualisation, Methodology, Investigation, Writing, review and editing of all versions of the article, Supervision, Project administration

Stéphane Bonny: Conceptualisation, Methodology, Validation, Formal Analysis, Visualisation, Writing, review and editing of all versions of the article

CRediT authorship contribution statement

Stéphane Bonny: Conceptualization, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Tereza Cahlikova:** Conceptualization, Investigation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

There are no competing interests to declare.

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